Design and Modeling of a Cable-Driven Rigid-Flexible Humanoid Hand

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***Abstract*—The research of humanoid multi-fingered hand aiming to be applied to people's daily life, but they are faced with the main contradiction between simplicity and functionality. Herein, we propose a novel rigid-flexible coupling humanoid multi-fingered hand comprising finger and palm. Based on the rigid-flexible coupling finger designed in this paper, the mechanical deformation of rigid-flexible coupling finger structure is analyzed from the static point of view based on the theory of beam constraint model. The mechanic model of the rigid-flexible coupling finger is established according to the deformation of the designed rigid-flexible coupling finger structure driven by the cable, considering the friction force of the cable. The feasibility of the design method is proved by experiments, which can potentially bean alternative for manipulators in practical applications to complete the flexible works.**

***Index Terms*—Humanoid hand, rigid-flexible coupling finger, cable-driven, mechanic modeling**

I. Introduction

Robots need dexterous hands to perform a variety of tasks. The main structure of traditional humanoid multi-fingered hand is mostly rigid. Traditional motors, cylinders and hydraulic cylinders are used as drivers, and a series of complex transmission system such as gears, belt wheels, connecting rods and transmission shafts are used to exert driving force on the humanoid knuckles, so as to realize long-distance driving of the knuckles. Humanoid multi-fingered hands with rigid structure can imitate the shape, structure and function of human hands. Typical examples of this kind of imitation hands include Shadow Hand[1], iClud Hand[2], DLR Hand[3] and DLR/HIT Hand II[4]. Although the multi-fingered hand of this kind of rigid structure is powerful, it has the disadvantages of complex structure, poor adaptability to the external environment and low safety, and it often requires complex drive control in practical applications.

At present, there are still some contradictions between the research design and practical application of humanoid multi-fingered hand, including the contradiction between the simplicity and dexterity of the imitation of human hand, and the contradiction between the rigid and the flexible characteristics of the production material.

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In response to the above two problems, many researchers began to propose a mechanical structure combining rigid and soft elastomers, which can make a good trade-off between rigid materials and flexible materials. In the literature, a number of rigid-flexible coupling technologies have been proposed. For example, a rigid-flexible coupling finger consisting of flexible joints coupled with rigid knuckles is proposed[5–7]. To reduce the flexural stiffness of flexible joints, the researchers compared the flexible joints of different structures[8], and finally chose the rectangular and wavy circular flexible joint as the joint of the finger. However, there are shortcomings in material selection and structure optimization, and the fingers with flexible joints cannot be deformed greatly. In addition, another pneumatic soft actuator with stiffening stiffness has been proposed[9–11]. This method can realize the stiffness strengthening of the soft structure within a certain stiffness variation range. By constraining the structure of the packed cavity particles and changing the blocking pressure mode, a kind of soft actuator with the stiffness strengthening of the chained particle layer is proposed[12]. The stiffness adjustment of the soft structure can be realized by pulling the chained particle structure by driving the cable.

In the previous research description of the rigid-flexible coupling structure, researchers basically focused on the idea of proposing a novel design method for the rigid-flexible coupling structure, and carried out simple static analysis and simulation verification using commercial finite element software for the proposed coupled structure, without engineering analysis of the theoretical model of the rigid-flexible coupling structure combined with the actual practice. For example, The rigid-flexible coupling hand in literature [6] is more concerned with the analysis model established between its underdriven structural characteristics and the grasping object.

In this paper, from the perspective of bionic design of humanoid multi-fingered hand, the design of rigid-flexible coupling humanoid multi-fingered hand structure, combined with the relevant theoretical studies of rigid and flexible structures, analyzes the interaction relationship of rigid-flexible coupling structures, can promote the development of research and design of humanoid multi-fingered hand, so as to promote the research and design of humanoid multi-fingered hand to practical application.

II. Design of the Rigid-flexible Humanoid Hand

A. Design of The Fingers

Fig. 1(a) shows a 3D model of the index finger which is mainly composed of the rigid material and the flexible

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material. The whole finger is formed by combining the flexible joints with the rigid phalanges. The size of the finger is designed according to the human hand, and the length of the finger joint is between five and six millimeter. The flexion movement of the humanoid finger is completed by the motor of the palm finger base driven by the cable, with 1 degree of freedom. The extension of the fingers is passively facilitated by the springback action of various flexible joints.

To facilitate easy assembly of the modular structure, the index finger, middle finger, and ring finger in this paper share identical shapes and sizes. The tail finger is slightly smaller in both shape and size. The thumb, given its crucial role in the human hand, is endowed with two degrees of freedom in this design. This enables the simulation of both the flexion movement of the thumb and the carpometacarpal joint (CMC joint), as shown in Fig. 1(b).

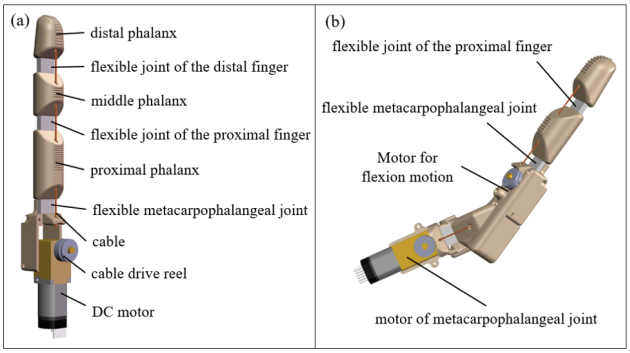


Fig. 1. Schematic structure of the rigid-flexible fingers, including the index finger (a) and the thumb finger (b).

In conjunction with the previously proposed design approach for the rigid-flexible humanoid finger structure, the finger in this configured structure is crafted by blending two materials using 3D printing molding technology, which is convenient to manufacture. Specifically, the material used in the rigid structure can be a high-stiffness 3D printing plastic which is not easy to deform. In this design and manufacture, a 3D printable thermoplastic polylactic acid material with high stiffness is used. Conversely, the material used in the flexible structure is a 3D printable thermoplastic polyurethane material with high elasticity and high recoverability. This material possesses a Shore hardness of 85 A, and its elastic modulus falls within the range of 20-26 MPa.

B. Palm Design and The Whole Hand

Fig. 2 illustrates the structural diagram of the palm section of the rigid-flexible humanoid hand. The design of the humanoid palm structure includes the base of palm, front shell of the palm, and back shell of the palm. The configuration of the base of the palm adheres to the positional layout principles observed in the actual human hand, guiding the placement of each finger within the palm.

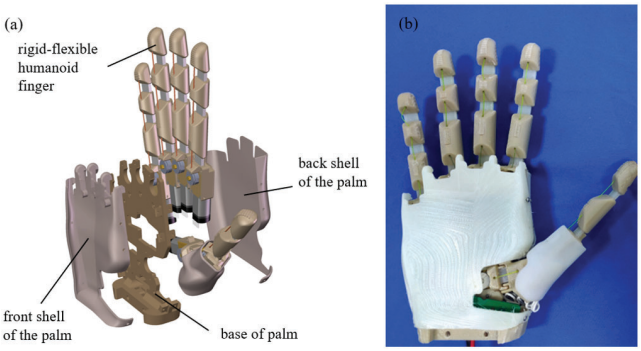


Fig. 2. The palm of the humanoid multi-fingered hand. (a) 3D model; (b) actual assembly illustration.

C. Control System Integration

The presented humanoid hand, detailed in this study, constitutes a tightly integrated system, necessitating the internal integration of the hardware control component within the palm. The circuit board's dimensions closely match those of the palm and are affixed to its interior. The control system, illustrated in Fig. 3, orchestrates the movement of six DC gear motors through a dedicated drive module. A sensor module serves to discern motor speed, position, and current, with subsequent data processing conducted by a microcontroller. Concurrently, to facilitate interactive control of the humanoid hand, we developed host-computer software. This software enables communication between the host computer and the microcontroller, allowing for the display of sensor information pertaining to each finger's movement and the transmission of instructions to govern the multi-fingered humanoid hand.

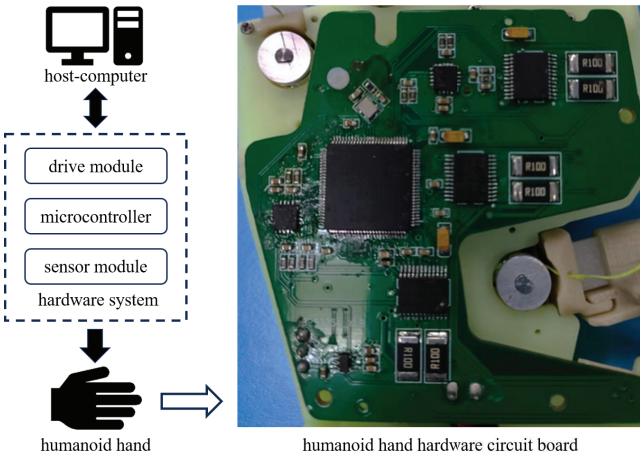


Fig. 3. Control system of the humanoid hand.

III. Modeling and Analysis

Our aim is to develop a rigid-flexible humanoid hand capable of executing human-like actions. Diverging from the conventional rigid multi-fingered hand, these actions are achieved through substantial deformations of the flexible joint within the

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rigid-flexible finger structure. In this section, employing the beam constraint model as a foundation, we utilize the chained Timoshenko beam constraint model. This model is adept at analyzing significant deformations in beams, providing a suitable framework for representing the rigid-flexible finger structure.

A. Mechanic Modeling

In conjunction with the previously described rigid-flexible humanoid structure, this section introduces a mathematical model for the force-deformation characteristics of a rigid-flexible coupling finger structure. The model is established based on the chained Timoshenko beam constraint model, analyzed statically. The rigid-flexible coupling structure proposed in this study diverges from the chained beam structure. In the chained beam structure, every component is a deformable beam element. In contrast, our design incorporates a rigid finger bone between each pair of beam elements, effectively introducing a rigid body element into the structure. This distinction means that the rigid-flexible coupling structure can be partitioned into multiple interconnected parts, forming a chain. Each part comprises a beam element and a rigid body element, as illustrated in Fig. 4.

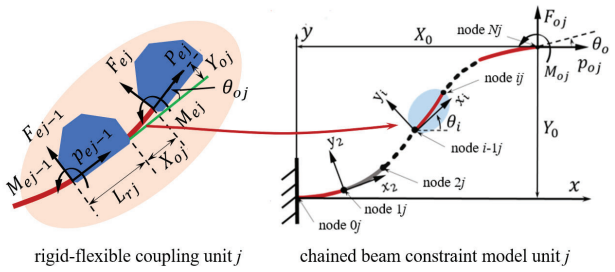


Fig. 4. Chained beam constraint model of the rigid-flexible unit.

The model for the rigid-flexible finger structure involves discretizing the finger structure into N_f coupling units connected end to end. The soft units within each rigid-flexible coupling unit are characterized using the chained Timoshenko beam constraint model. For the j-th rigid-flexible coupling element $(2 \leq j \leq N_f)$, a local coordinate system $(O_j X_j Y_j)$ is established along the tangential direction of the end of the $(j-1)$-th element and moves with it. The local coordinate system for the first unit is fixed at the rigid-flexible coupling finger structure's fixed end (node 0), and the finger's termination is represented by node N_f. In accordance with the chained Timoshenko beam constraint model[13,14], the axial load, lateral load, and bending moment on node j in the local coordinate system of the j-th element are denoted as P_{oj}, F_{oj} and M_{oj}, respectively. When the j-th rigid-flexible coupling element rotates by an angle θ_{oj} around the $(j-1)$-th rigid-flexible coupling element, a static equilibrium equation holds:

$$\begin{cases} f_{oj} = \frac{F_{oj}L^2}{EI} = N^2 f_{1j} \\ p_{oj} = \frac{P_{oj}L^2}{EI} = N^2 p_{1j} \\ m_{oj} = \frac{M_{oj}L^2}{EI} = N m_{Nj} \end{cases} \tag{1}$$

$$p_{oj} = \sum_{1}^{i} \alpha_{ji} \tag{2}$$

$$\begin{bmatrix} P_{oj} \\ F_{oj} \\ M_{oj} \end{bmatrix} = \begin{bmatrix} \cos\theta_{oj} & -\sin\theta_{oj} & 0 \\ \sin\theta_{oj} & \cos\theta_{oj} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} P_{ej} \\ F_{ej} \\ M_{ej} \end{bmatrix} \tag{3}$$

Hence, when the three parameters defining the end load are specified for the rigid body element within the rigid-flexible coupling section, the deformation of the beam element in the j-th rigid-flexible coupling element can be determined. This is achieved by incorporating the aforementioned chained Timoshenko beam constraint model. Subsequently, the deformation extends through the beam elements of the $j-1$, $j-2$, and so forth, until reaching the first rigid-flexible coupling element. In this manner, the mathematical model for the rigid-flexible coupling finger can be systematically solved under varying loads.

B. Force Analysis

The designed rigid-flexible finger in this study aims to achieve the flexion movement of a human finger through cable driving. Therefore, it is imperative to account for the tangible impact of cables on the rigid-flexible finger structure. This entails combining the previously mentioned mathematical model of the rigid-flexible finger with an analysis of the forces acting on the structure under cable driving conditions. In the subsequent discussion, we illustrate the application of this approach using a rigid-flexible finger with three knuckles as an example.

Considering cable friction, the cable holes in each rigid finger bone of the finger structure are continually influenced by the cable force. For detailed analysis, the j-th rigid-flexible coupling element is selected, and the cable interacts with the rigid phalanges at the cable holes situated at the left and right ends, as depicted in Fig. 5(a).

During this scenario, the cable adheres to a static balance relationship at the entry and exit ends of the rigid finger bone:

$$\begin{cases} F_j = F_{jl} + F_{jr} \\ N_{jl} = N_{jl} + f_{jl} = T_{jl} + T_{jinl} \\ F_{jr} = N_{jr} + f_{jr} = T_{jinr} + T_{jr} \end{cases} \tag{4}$$

In this equilibrium relationship, F_j represents the combined force exerted by the left and right ends of the rigid phalanges on the cable. Additionally, F_{jl} and F_{jr} denote the forces applied by the left and right ends of the rigid phalanges on the cable, respectively.

To determine the entry and exit direction of the cable at the left end of the finger bone, several factors are considered. These include the extension position of the cable line at the right end of the $(j-1)$-th finger bone, the deformation of the beam element within the j-th rigid and flexible coupling section, and the orientation transformation matrix H_{ljr} from the end of the beam element to the entry position at the left end of the j-th finger bone. Conversely, the direction of the cable extension at the right end of the finger bone is determined

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by the extension position of the cable at the right end of the j-th finger bone, the deformation of the beam element within the $(j+1)$-th rigid-flexible coupling section, and the uniform transformation matrix H_{rjl}. This matrix describes the entry position from the end of the $(j+1)$-th beam element to the left end of the $(j+1)$-th finger bone. The expressions for H_{ljr} and H_{rjl}, in conjunction with Fig. 5(b), are outlined as follows:

$$\begin{cases} H_{ljr} = H_{lej} H_{ebj} H_{bjr} \\ H_{rjl} = H_{rb(j+1)} H_{b(j+1)e} H_{e(j+1)l} \end{cases} \tag{5}$$

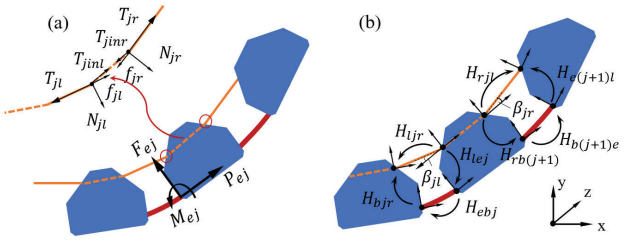


Fig. 5. Static analysis. (a) Force analysis of the cable in the finger; (b) Analysis of the direction of the incoming and outgoing ends of the cable.

By integrating all the aforementioned formulas with the cable winch equation[15], and subsequently simplifying the derivation, it can be established that when the force exerted by the cable on the rigid finger bone is transmitted to the end of the beam element within the j-th rigid-flexible coupling element, the resulting moment, denoted as M_{ej}, is obtained.

$$M_{ej} = F_{j1} \times r_l + F_{jr} \times r_r \tag{6}$$

By amalgamating the preceding content, the stress, and deformation characteristics of the rigid-flexible coupling finger under the influence of cable action can be determined. In conjunction with the mathematical model presented in the previous section of this paper, the static mathematical model for the finger structure comprising N_{unit} rigid-flexible coupling units under cable driving conditions can be derived. Fig. 7 illustrates the static solution of the mathematical model for N_{unit} values of 1, 2, and 3. Moreover, the friction coefficient between the cable and the rigid finger bone, a critical factor in the mathematical model solution, was empirically measured as 0.25[15].

C. Simulation Verification

In this section, the rigid-flexible coupling finger structure is simulated and validated utilizing the Abaqus finite element software. The simulation involves conducting finite element simulation calculations for the rigid-flexible coupling finger structure, specifically focusing on configurations with 1-3 rigid-flexible coupling elements. These simulations are performed under the influence of varying cable driving forces. The corresponding results of the simulations are depicted in Fig. 7.

D. Experimental Verification

After conducting the finite element simulation calculation, it is imperative to further validate the accuracy of the proposed mathematical model by integrating experimental data. Consequently, the movement of the rigid-flexible humanoid finger structure driven by a cable is systematically compared and verified from an experimental standpoint. The experimental platform, depicted in Fig. 6, is established for this purpose.

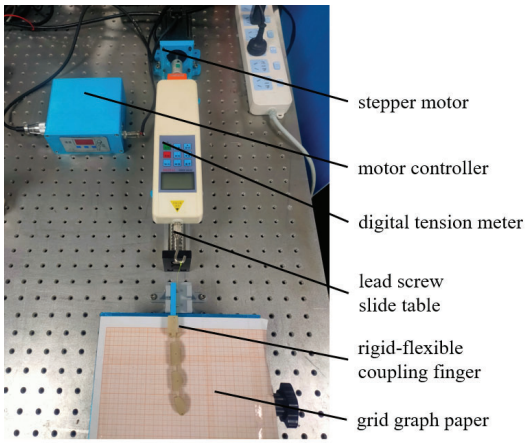


Fig. 6. Cable-driven rigid-flexible fingers experimental platform.

During the experiment, the rigid-flexible fingers with different number of phalanges were fixed on the bracket, one end of the cable was connected to the end of the finger, and the other end was connected to the digital display push tension meter. The digital display push tension meter is fixed on the lead screw slide, and the step motor rotates slowly and evenly to make the lead screw slide move and drive the cable to produce tension and displacement. The cable will drive the rigid-flexible coupling finger structure to deform, and the digital push tension meter will record the tension value of the cable in real time. The deformation movement information of the rigid-flexible coupled finger structure is obtained by setting mark points at the finger tip of the finger structure and placing coordinate paper near the rigid-flexible coupled finger, so that the camera perpendicular to the rigid-flexible finger is recorded and measured.

To minimize experimental error, each set of experiments was repeated three times and the average value was calculated. Among them, each group was loaded with a cable driving force ranging from 0 N to 3 N, each loading interval was 0.2 N, and each group was loaded 15 times to obtain the fingertip marker position data with different rigid-flexible coupling elements under different cable driving forces. After sorting out these experimental data, combined with the above mentioned mathematical model and finite element simulation calculation, the data comparison diagram of the three can be drawn, as shown in Fig. 7.

By analyzing the distance error of the finger end point, the maximum error of comparison between modeling and experimental data with different rigid-flexible coupling ele-

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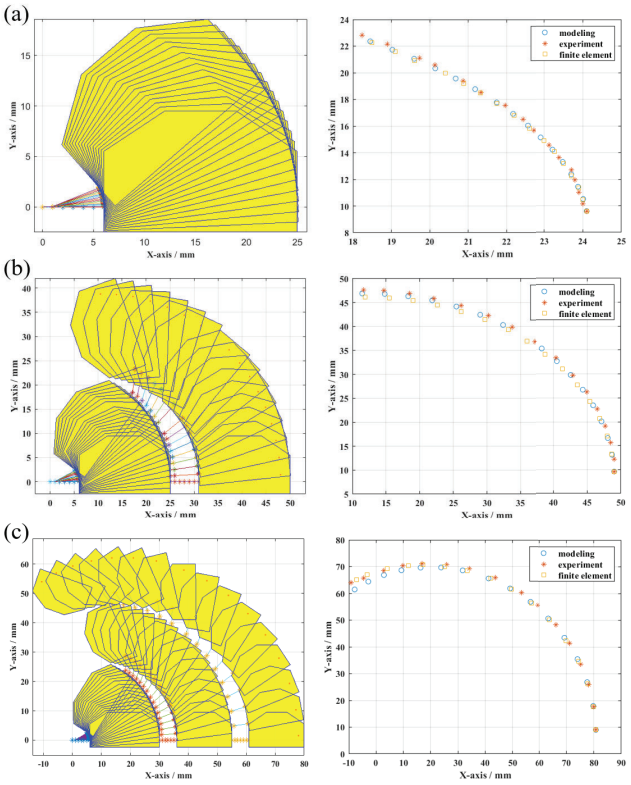


Fig. 7. Comparison of force-deformation data for different number of rigid-flexible coupling units. (a) one unit; (b) two units; (c) three units.

ments is 4.4 mm. As the number of rigid-flexible coupling elements increases, the error of modeling calculation gradually increases, and the average error between modeling data and experimental comparison with three-unit calculation is 2.2 mm. Compared with the finite element error, it can be seen that the error size and trend change of the established mathematical model are basically similar to that of the simulation data. The main sources of these errors are: (1) Flexible joints are affected by the nonlinear hyperelasticity of flexible materials, and linear elastomers are used in modeling and finite element calculation to simplify the constitutive equations of this material. (2) The nonlinear friction generated by the action of the rigid unit during the cable drive process is also simplified in the calculation process. (3) The instability of 3D printing manufacturing process. Since the rigid-flexible coupling finger structure is manufactured by 3D printing deposition principle, the material characteristics of the manufactured flexible joint will be lower than those of traditional manufacturing methods. The material pcablerties of flexible joints are changed. Therefore, based on the experimental data, the mathematical model of the rigid-flexible coupling finger structure established by different rigid-flexible coupling elements is compared with the finite element calculation data, and it is proved that the mathematical model of the rigid-flexible coupling finger structure driven by the cable has high precision and good accuracy.

IV. Tests and Experiments

A. Basic Performance Tests

The rigid-flexible coupling finger structure performance is related to the use of the cable driven DC gear motor. In this paper, the cable-driven DC gear motor operates at a load speed of 65 RPM, with an average torque of 0.3 N·m under a working voltage of 12 V and a blocked current of 0.5 A. Test results indicate that the manufactured rigid-flexible coupled finger structure requires a maximum flexural tension of approximately 8.6 N when undergoing the most extensive flexural motion. Additionally, a fundamental performance test is conducted by employing the cable to drive the DC gear motor, propelling the rigid-flexible coupling finger structure. The maximum flexion motion of the rigid-flexible coupling finger structure is illustrated in Fig. 8(a). During this state, the near flexible joints, medium flexible joints, and far flexible joints of the finger structure can achieve a bending angle exceeding 90 degrees.

Furthermore, the maximum pressing force of the rigid-flexible coupling finger structure is assessed using a digital display tension gauge, as depicted in Fig. 8(b). In instances where the flexible joints of the rigid-flexible coupling finger structure are in the initial bending state with only slight bending deformation, the DC deceleration motor imparts pressure at the end of the finger through the cable. After five tests, the average pressure is approximately 15.6 N.

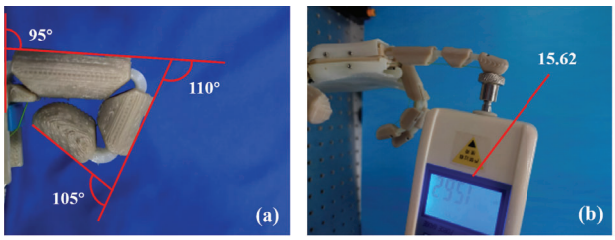


Fig. 8. Performance test of the rigid-flexible finger. (a) Maximum flexion test; (b) Press force test.

B. Grasping Experiments

In this study, a grasping experiment is carried out on a humanoid hand. To increase friction, gloves were applied to the humanoid hand before the experiment. Fig. 9 illustrates a comparison between the gloved prosthetic hand and the human hand. As can be seen from Fig. 10, the rigid-flexible humanoid hand designed in this paper can basically complete various grasping tests of grasping classification, so the rigid-flexible humanoid hand can imitate the grasping function of human hands to a certain extent, and has the ability to complete the grasping of daily objects.

V. Conclusion

In this paper, a modular rigid-flexible coupling humanoid hand which can be manufactured in one piece is designed. The size of the humanoid hand is designed according to the real human hand. Then, based on the theory of beam constraint

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Fig. 9. The comparative illustration between the gloved humanoid hand (left) and the human hand (right).



Fig. 10. Grasping experiments with the humanoid hand.

model, a mechanic model of the chained rigid-flexible finger structure is established. In the case of friction, the rigid-flexible finger structure driven by cable is analyzed and its mechanic model is established. To validate the accuracy of the mechanic model for the cable-driven rigid-flexible finger structure, a series of finite element simulation calculations and experiments are executed. The outcomes affirm the reliability of the proposed mechanic model. Finally, comprehensive performance tests and experiments for the rigid-flexible humanoid hand are undertaken. These include assessments of the maximum flexion amplitude, maximum flexion force, and a grasping experiment. The collective results substantiate the feasibility and effectiveness of the proposed design method for the rigid-flexible finger structure.

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